

Акторы (People)

Person

```
- smoke_damage_per_step: int = 2
- max_smoke_exposure: int = 25
- fire_damage_per_step: int = 3
- max_fire_exposure: int = 18
+ id: int
+ x: int
+ y: int
+ isAlive: boolean
+ smokeExposure: int
+ fireExposure: int
+ targetExitX: int
+ targetExitY: int
+ isFollowingExitSign: boolean
+ isEvacuating: boolean
+ hasPressedAlarm: boolean
+ evacuationStartTime: int
+ Person(id: int, startX: int, startY: int)
+ increaseSmokeExposure(): void
+ increaseFireExposure(): void
+ startEvacuation(currentTimeStep: int): void
+ getHealthPercentage(): double
+ canTakeFireExtinguisher(): boolean
+ canPickUpFireExtinguisher(): void
+ canFightFire(): boolean
+ onFightFire(sim: Simulation): void
+ getRoleName(): String
+ hasFireExtinguisher(): boolean
+ isHero(): boolean
```

Civilian

```
+ Civilian(id: int, startX: int, startY: int)
+ canTakeFireExtinguisher(): boolean
+ onPickUpFireExtinguisher(): void
+ canFightFire(): boolean
+ onFightFire(sim: Simulation): void
+ getRoleName(): String
+ hasFireExtinguisher(): boolean
```

Hero

```
+ hasFireExtinguisher: boolean
+ Hero(id: int, startX: int, startY: int)
+ hasFireExtinguisher(): boolean
+ pickUpFireExtinguisher(): void
+ useFireExtinguisher(): void
+ canTakeFireExtinguisher(): boolean
+ onPickUpFireExtinguisher(): void
+ canFightFire(): boolean
+ onFightFire(sim: Simulation): void
+ getRoleName(): String
```

Пожар (Fire)

Fire

```
- initial_intensity: int = 5
- max_intensity: int = 30
- base_spread_probability: double = 0.30
- spread_intensity_bonus: double = 0.05
- max_spread_probability: double = 0.80
- base_smoke_probability: double = 0.30
- smoke_intensity_bonus: double = 0.03
+ x: int
+ y: int
+ intensity: int
+ Fire(x: int, y: int)
+ spread(): void
+ getIntensity(): int
+ getSpreadProbability(): double
+ getSmokeProbability(): double
+ getDamagePerStep(): int
+ toString(): String
```

Симуляция (Simulation)

Simulation

```
- time_between_people_min: int = 5
- time_between_people_max: int = 20
- hero_spawn_chance_percent: int = 15
- fire_spread_fire_chance_percent: int = 35
- fire_spread_smoke_chance_percent: int = 20
- spread_interval_steps: int = 4
- forced_fire_step: int = 80
- fire_start_threshold_step_1: int = 30
- fire_start_chance_1: int = 35
- fire_start_threshold_step_2: int = 40
- fire_start_chance_2: int = 30
- fire_start_threshold_step_3: int = 50
- fire_start_chance_3: int = 50
- fire_start_threshold_step_4: int = 60
- fire_start_chance_4: int = 80
- random_range_max: int = 100
- animation_delay_ms: int = 200
+ building: Building
+ people: ArrayList<Person>
+ fires: ArrayList<Fire>
+ fireStarted: boolean
+ timeStep: int
+ alarmPressed: int
+ random: Random
+ nextPersonId: int
+ timeUntilNextPerson: int
+ totalPeopleEver: int
+ totalHeroesEver: int
+ totalCiviliansEver: int
+ spreadCount: int
+ fireStartTime: int
+ evacuatedCount: int
+ buildingName: String
+ heroesPressedAlarmTotal: int
+ civiliansPressedAlarmTotal: int
+ Simulation(width: int, height: int)
+ addPerson(isHero: boolean): void
+ startFire(): void
+ spreadFire(): void
+ findNearestExit(x: int, y: int): Point
+ findNearestButton(x: int, y: int): Point
+ hasUnpressedButtons(): boolean
+ movePerson(p: Person): void
+ updatePeople(): void
+ addRandomPerson(): void
+ startFireRandomly(): void
+ steps(): void
+ run(steps: int): void
```

Помещение (Building)

Building

```
+ width: int
+ height: int
+ roomId: RoomCellID
+ exits: ArrayList<Point>
+ windows: ArrayList<Point>
+ fireExtinguishers: ArrayList<Point>
+ alarmButtons: ArrayList<Point>
+ exitSigns: ArrayList<Point>
+ Building(width: int, height: int)
+ clearAll(): void
+ createWalls(): void
+ loadFromFile(fileName: String): void
+ addExit(x: int, y: int): void
+ addWindow(x: int, y: int): void
+ addFireExtinguisher(x: int, y: int): void
+ addAlarmButton(x: int, y: int): void
+ addExitSign(x: int, y: int): void
+ setFire(x: int, y: int): void
+ setSmoke(x: int, y: int): void
+ clearFireAndSmoke(): void
+ isInside(x: int, y: int): boolean
+ isWalkable(x: int, y: int): boolean
+ printMap(): void
```

Direction

```
+ up: int
+ right: int
+ down: int
+ left: int
+ none: int
+ all: int[]
+ getDx(direction: int): int
+ getDy(direction: int): int
+ isValid(direction: int): boolean
```

Пользовательский интерфейс (GUI)

StatisticsWindow

```
- window_width: int = 1000
- window_height: int = 800
- report_width: int = 750
- report_height: int = 750
- map_text_cols: int = 35
- map_text_rows: int = 25
- stats_text_cols: int = 35
- stats_text_rows: int = 20
- people_text_cols: int = 32
- people_text_rows: int = 12
- health_bar_length: int = 10
- fire_exposure_max: int = 18
- smoke_exposure_max: int = 25
+ mapArea: JTextArea
+ statsArea: JTextArea
+ peopleArea: JTextArea
+ stepLabel: JLabel
+ animationPanel: AnimationPanel
+ StatisticsWindow()
+ showAlarmAnimation(): void
+ updateDisplay(sim: Simulation, totalPeopleEver: int, totalHeroesEver: int, totalCiviliansEver: int, spreadCount: int, fireStartTime: int): void
+ buildMap(sim: Simulation): String
+ buildStats(sim: Simulation): String
+ buildPeopleStatus(sim: Simulation): String
+ getSymbol(sim: Simulation, x: int, y: int): char
+ getHealthBar(percent: int): String
```

AnimationPanel

```
- animation_delay_ms: int = 50
- flame_count: int = 20
- min_flame_height: int = 10
- max_flame_height: int = 60
- min_flame_width: int = 8
- max_flame_width: int = 20
- alarm_animation_duration: int = 20
- alarm_blink_interval: int = 4
- smoke_layers: int = 10
- smoke_base_alpha: int = 40
- max_smoke_alpha: int = 80
+ currentSim: Simulation
+ frameCounter: int
+ frameHeights: int[]
+ frameWidths: int[]
+ alarmAnimation: boolean
+ alarmAnimationCounter: int
+ AnimationPanel()
+ updateFireData(sim: Simulation): void
+ startAlarmAnimation(): void
+ countFireCells(): int
+ countSmokeCells(): int
+ paintComponent(g: Graphics): void
```

RoomCell

```
+ type_empty: int = 0
+ type_wall: int = 1
+ type_exit: int = 2
+ type_window: int = 3
+ type_fire: int = 4
+ type_smoke: int = 5
+ type: int
+ hasFireExtinguisher: boolean
+ hasAlarmButton: boolean
+ hasExitSign: boolean
+ RoomCell()
+ RoomCell(type: int)
+ isWalkable(): boolean
+ isFire(): boolean
+ isSmoker: boolean
+ isExit(): boolean
+ isWall(): boolean
+ isEmpty(): boolean
+ isWindow(): boolean
+ getSymbol(): char
+ toString(): String
```

Point

```
+ x: int
+ y: int
+ Point(x: int, y: int)
+ Point(other: Point)
+ equals(obj: Object): boolean
+ getDistance(targetX: int, targetY: int): int
+ toString(): String
```

ArrayList

```
+ size(): int
+ add(element: T): void
+ get(index: int): T
+ remove(index: int): void
+ clear(): void
+ isEmpty(): boolean
```

Random

```
+ nextInt(bound: int): int
+ nextInt(): int
+ nextDouble(): double
```

Бенчмарк (Benchmarks)